

Practice Questions

Physical Sciences — Paper 1 | Week 1: Scientific Notation, SI Units & Prefixes

SECTION A: Scientific Notation (10 questions)

Convert to scientific notation:

1. 450 000 000
2. 0.000 000 072
3. 3 200 000 000 000
4. 0.000 45
5. 93 000 000

Convert from scientific notation to ordinary numbers:

6. 6.02×10^{23}
7. 3.4×10^{-5}
8. 1.67×10^{-27}
9. 9.1×10^6
10. 2.998×10^8

SECTION B: SI Prefix Conversions (10 questions)

11. Convert 5 km to metres
12. Convert 350 mm to metres
13. Convert 0.0025 m to millimetres
14. Convert 8 500 000 Hz to megahertz (MHz)
15. Convert 3.2 gigabytes (GB) to bytes
16. Convert 450 nanometres (nm) to metres
17. Convert 0.000 062 metres to micrometres (μm)
18. Convert 7 500 milligrams (mg) to kilograms (kg)
19. Convert 2.4×10^{-3} km to millimetres
20. A bacterium is 2 μm long. Express this in metres using scientific notation

ANSWER KEY

Q	Answer	Q	Answer
1	4.5×10^8	11	5 000 m
2	7.2×10^{-8}	12	0.350 m
3	3.2×10^{12}	13	2.5 mm
4	4.5×10^{-4}	14	8.5 MHz
5	9.3×10^7	15	3 200 000 000 bytes
6	602 000 000 000 000 000 000 000	16	4.5×10^{-7} m
7	0.000 034	17	62 μ m
8	0.000 000 000 000 000 000 000 001 67	18	0.0075 kg
9	9 100 000	19	2 400 mm
10	299 800 000	20	2×10^{-6} m

PREFIX REFERENCE CARD

Prefix	Symbol	Multiply by
giga	G	10^9
mega	M	10^6
kilo	k	10^3
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}

Memorise this table. Every prefix question is just multiplying or dividing by the correct power of 10.